

ASSIGNMENT-13.4

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BATCH: 46

Task Description #1 (Refactoring – Removing Global Variables)

- **Task:** Use AI to eliminate unnecessary global variables from the code.

- **Instructions:**

- o Identify global variables used across functions.

- o Refactor the code to pass values using function parameters.

- o Improve modularity and testability.

- **Sample Legacy Code:**

```
rate = 0.1
```

```
def calculate_interest(amount):
```

```
    return amount * rate
```

```
print(calculate_interest(1000))
```

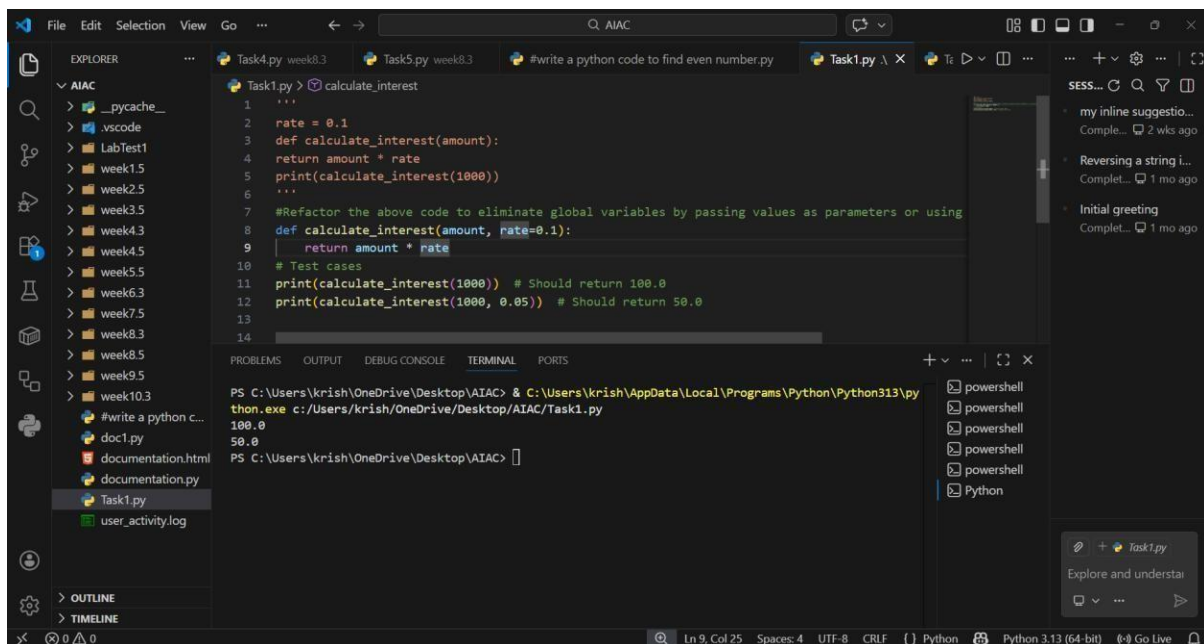
- **Expected Output:**

- o Refactored version passing rate as a parameter or using a configuration structure.

PROMPT:

“Refactor the above code to eliminate global variables by passing values as parameters or using configuration objects.”

CODE AND OUTPUT:



The screenshot shows a VS Code editor with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer shows a project named 'AIAC' with various files and folders. The code editor displays a Python script named 'Task1.py' with the following code:

```
1 '''
2 rate = 0.1
3 def calculate_interest(amount):
4     return amount * rate
5     print(calculate_interest(1000))
6 '''
7 #Refactor the above code to eliminate global variables by passing values as parameters or using
8 def calculate_interest(amount, rate=0.1):
9     return amount * rate
10 # Test cases
11 print(calculate_interest(1000)) # Should return 100.0
12 print(calculate_interest(1000, 0.05)) # Should return 50.0
13
14
```

The terminal at the bottom shows the command to run the script and its output:

```
PS C:\Users\krish\OneDrive\Desktop\AIAC> & C:\Users\krish\AppData\Local\Programs\Python\Python313\python.exe c:/Users/krish/OneDrive/Desktop/AIAC/Task1.py
100.0
50.0
PS C:\Users\krish\OneDrive\Desktop\AIAC>
```

EXPLANATION:

This ensures modularity and testability by avoiding hidden dependencies. Functions become reusable since they rely only on inputs, not external state. It improves clarity and makes unit testing easier.

Task Description #2 : (Refactoring Deeply Nested Conditionals)

- **Task:** Use AI to refactor deeply nested if–elif–else logic into a cleaner structure.

- **Focus Areas:**

- o Readability

- o Logical simplification

- o Maintainability

Legacy Code:

score = 78

if score >= 90:

print("Excellent")

else:

```
if score >= 75:
```

```
    print("Very Good")
```

```
else:
```

```
    if score >= 60:
```

```
        print("Good")
```

```
    else:
```

```
        print("Needs Improvement")
```

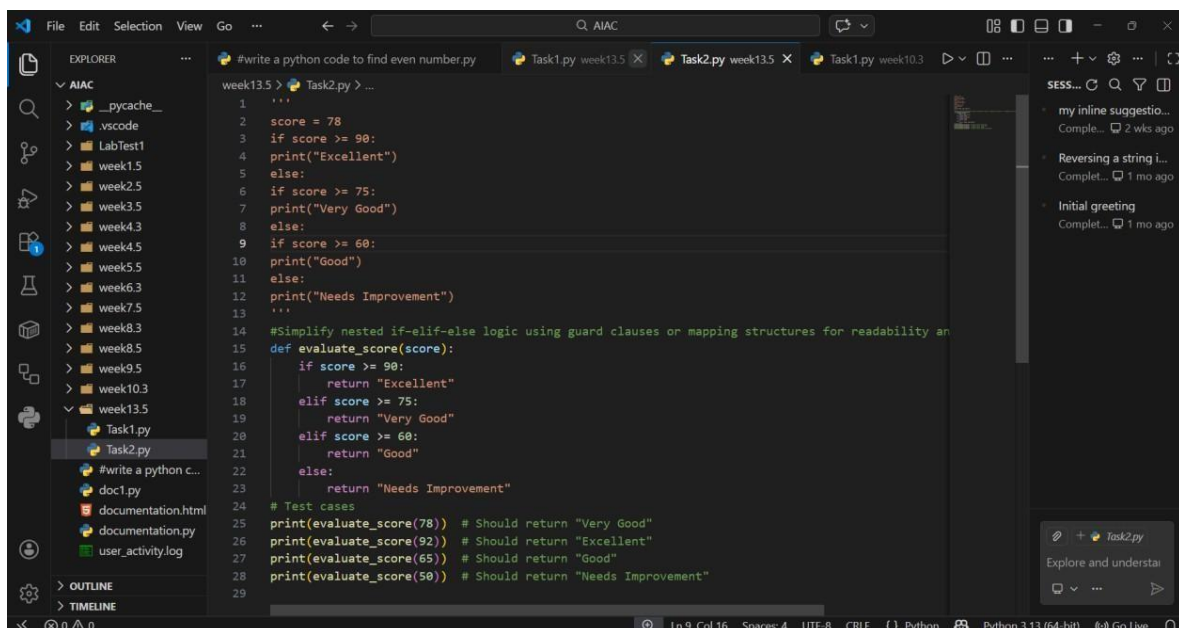
Expected Outcome:

o Flattened logic using guard clauses or a mapping-based approach.

PROMPT:

“Simplify nested if–elif–else logic using guard clauses or mapping structures for readability and maintainability.”

CODE:



```
#write a python code to find even number.py
1  '''
2  score = 78
3  if score >= 90:
4  print("Excellent")
5  else:
6  if score >= 75:
7  print("Very Good")
8  else:
9  if score >= 60:
10 print("Good")
11 else:
12 print("Needs Improvement")
13 '''
14 #Simplify nested if-elif-else logic using guard clauses or mapping structures for readability and maintainability
15 def evaluate_score(score):
16     if score >= 90:
17         return "Excellent"
18     elif score >= 75:
19         return "Very Good"
20     elif score >= 60:
21         return "Good"
22     else:
23         return "Needs Improvement"
24 # Test cases
25 print(evaluate_score(78)) # Should return "Very Good"
26 print(evaluate_score(92)) # Should return "Excellent"
27 print(evaluate_score(65)) # Should return "Good"
28 print(evaluate_score(50)) # Should return "Needs Improvement"
29
```

OUTPUT:

```
thon.exe c:/Users/krish/OneDrive/Desktop/AIAC/week13.5/Task2.py
Very Good
Excellent
Good
Needs Improvement
PS C:\Users\krish\OneDrive\Desktop\AIAC>
```

EXPLANATION:

Flattening conditionals reduces cognitive load and makes logic easier to follow. Guard clauses exit early, while mappings provide a clean lookup. This improves readability and future scalability.

Task 3 (Refactoring Repeated File Handling Code)

- **Task:** Use AI to refactor repeated file open/read/close logic.

- **Focus Areas:**

- o DRY principle

- o Context managers

- o Function reuse

Legacy Code:

```
f = open("data1.txt")
```

```
print(f.read())
```

```
f.close()
```

```
f = open("data2.txt")
```

```
print(f.read())
```

```
f.close()
```

Expected Outcome:

- o Reusable function using with open() and parameters.

PROMPT:

“Apply DRY principles by refactoring repeated file open/read/close code into a reusable function with context managers.”

CODE AND OUTPUT:

```
1  ...
2  f = open("data1.txt")
3  print(f.read())
4  f.close()
5  f = open("data2.txt")
6  print(f.read())
7  f.close()
8  ...
9  #Apply DRY principles by refactoring repeated file open/read/close code
10 def read_file(filename: str) -> str:
11     """Return the contents of *filename* or a descriptive error message.
12     The files are expected to be readable text files; caller may supply a
13     relative path. A FileNotFoundError is caught so the script doesn't c
14     """
15     try:
16         with open(filename, "r") as f:
17             return f.read()
18     except FileNotFoundError:
19         return f"Error: {filename!r} not found."
20     except OSError as exc:
21         return f"Error opening {filename!r}: {exc}"
22
23 print(read_file("data1.txt")) # adjust path if necessary
24 print(read_file("data2.txt")) # adjust path if necessary
25
```

EXPLANATION:

Using with open() ensures safe resource handling and avoids manual closing. Encapsulating file logic in a function reduces duplication. This makes the code concise, reusable, and less error-prone.

Task 4 (Optimizing Search Logic)

- Task: Refactor inefficient linear searches using appropriate data structures.

• Focus Areas:

- o Time complexity
- o Data structure choice

Legacy Code:

```
users = ["admin", "guest", "editor", "viewer"]
```

```
name = input("Enter username: ")
```

```
found = False
```

```
for u in users:
```

```
if u == name:
```

```
found = True
```

```
print("Access Granted" if found else "Access Denied")
```

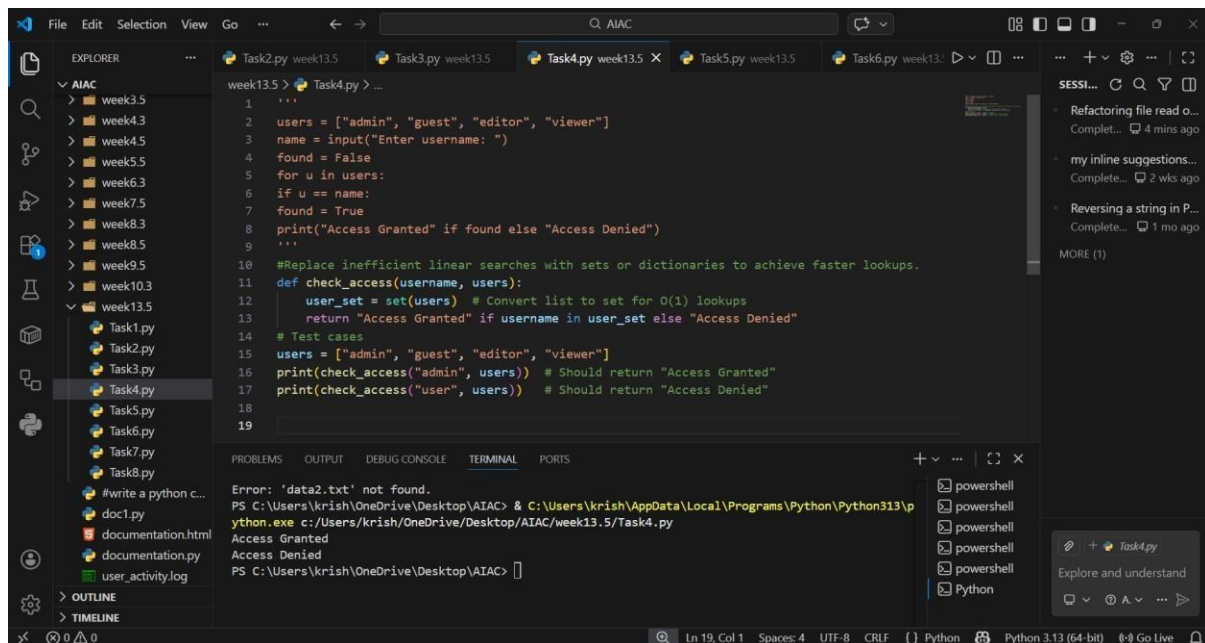
Expected Outcome:

- o Use of sets or dictionaries with complexity justification.

PROMPT:

“Replace inefficient linear searches with sets or dictionaries to achieve faster lookups.”

CODE AND OUTPUT:



The screenshot shows a VS Code editor with a file explorer on the left displaying a directory structure for 'AIAC' with subfolders 'week3.5' through 'week13.5' and files 'Task1.py' through 'Task8.py'. The main editor window shows a Python script 'Task4.py' with the following code:

```
1 ...
2 users = ["admin", "guest", "editor", "viewer"]
3 name = input("Enter username: ")
4 found = False
5 for u in users:
6     if u == name:
7         found = True
8     print("Access Granted" if found else "Access Denied")
9 ...
10 #Replace inefficient linear searches with sets or dictionaries to achieve faster lookups.
11 def check_access(username, users):
12     user_set = set(users) # Convert list to set for O(1) lookups
13     return "Access Granted" if username in user_set else "Access Denied"
14 # Test cases
15 users = ["admin", "guest", "editor", "viewer"]
16 print(check_access("admin", users)) # Should return "Access Granted"
17 print(check_access("user", users))  # Should return "Access Denied"
18
19
```

The bottom panel shows the 'TERMINAL' tab with the following output:

```
PS C:\Users\krish\OneDrive\Desktop\AIAC> & C:\Users\krish\AppData\Local\Programs\Python\Python313\python.exe c:/Users/krish/OneDrive/Desktop/AIAC/week13.5/Task4.py
Access Granted
Access Denied
PS C:\Users\krish\OneDrive\Desktop\AIAC>
```

EXPLANATION:

Linear search has $O(n)$ complexity, while sets/dictionaries provide average $O(1)$ lookups. This drastically improves performance for large datasets. It also makes the code cleaner and more efficient.

Task 5 (Refactoring Procedural Code into OOP Design)

- Task: Use AI to refactor procedural code into a class-based design.

- Focus Areas:

- o Object-Oriented principles

- o Encapsulation

Legacy Code:

```
salary = 50000
```

```
tax = salary * 0.2
```

```
net = salary - tax
```

```
print(net)
```

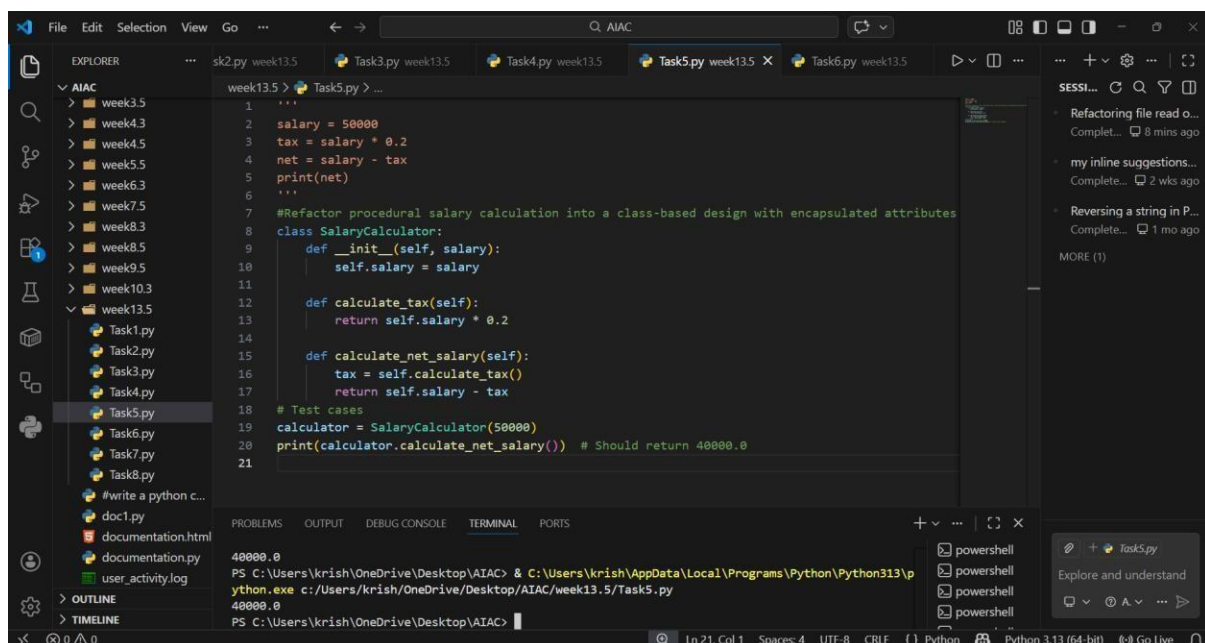
Expected Outcome:

o A class like `EmployeeSalaryCalculator` with methods and attributes.

PROMPT:

“Refactor procedural salary calculation into a class-based design with encapsulated attributes and methods.”

CODE AND OUTPUT:



The screenshot shows a Visual Studio Code editor with a Python file named `Task5.py` open. The code implements a `SalaryCalculator` class. The `__init__` method takes a `salary` parameter and stores it as `self.salary`. The `calculate_tax` method returns `self.salary * 0.2`. The `calculate_net_salary` method calls `calculate_tax` and returns `self.salary - tax`. Test cases are included at the bottom, creating a `SalaryCalculator` instance with a salary of 50000 and printing the result of `calculate_net_salary()`, which is 40000.0. The terminal at the bottom shows the execution of the script, outputting `40000.0` twice. The Explorer sidebar on the left shows a project structure with folders `AIAC` and `week13.5`, and several `Task` files. The bottom status bar indicates the file is at line 21, column 1, using UTF-8 encoding and CRLF line endings, with Python 3.13 (64-bit) selected.

```
1  """
2  salary = 50000
3  tax = salary * 0.2
4  net = salary - tax
5  print(net)
6  """
7  #Refactor procedural salary calculation into a class-based design with encapsulated attributes
8  class SalaryCalculator:
9      def __init__(self, salary):
10         self.salary = salary
11
12     def calculate_tax(self):
13         return self.salary * 0.2
14
15     def calculate_net_salary(self):
16         tax = self.calculate_tax()
17         return self.salary - tax
18
19     # Test cases
20     calculator = SalaryCalculator(50000)
21     print(calculator.calculate_net_salary()) # Should return 40000.0
```

```
40000.0
PS C:\Users\krish\OneDrive\Desktop\AIAC> & C:\Users\krish\AppData\Local\Programs\Python\Python313\python.exe c:/Users/krish/OneDrive/Desktop/AIAC/week13.5/Task5.py
40000.0
PS C:\Users\krish\OneDrive\Desktop\AIAC>
```

EXPLANATION:

Object-Oriented design improves structure and reusability. Encapsulation keeps salary, tax, and net logic within a class. This makes the code extensible, testable, and aligned with OOP principles.

Task 6 (Refactoring for Performance Optimization)

- Task: Use AI to refactor a performance-heavy loop handling

large data.

- Focus Areas:

- o Algorithmic optimization

- o Use of built-in functions

Legacy Code:

```
total = 0
```

```
for i in range(1, 1000000):
```

```
    if i % 2 == 0:
```

```
        total += i
```

```
print(total)
```

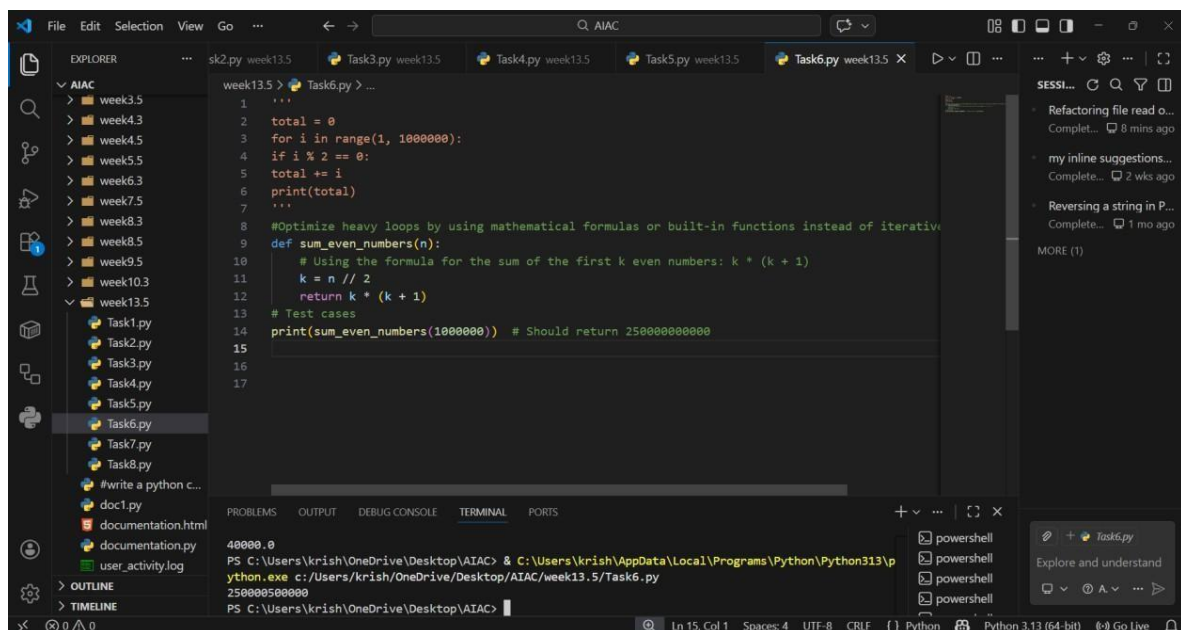
Expected Outcome:

- o Optimized logic using mathematical formulas or comprehensions, with time comparison.

PROMPT:

"Optimize heavy loops by using mathematical formulas or built-in functions instead of iterative summation."

CODE AND OUTPUT:



The screenshot shows a Visual Studio Code editor window with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer shows a project named 'AIAC' with several folders and files. The code editor displays a Python script named 'Task6.py' with the following code:

```
1 '''
2 total = 0
3 for i in range(1, 1000000):
4     if i % 2 == 0:
5         total += i
6     print(total)
7 '''
8 #Optimize heavy loops by using mathematical formulas or built-in functions instead of iterative
9 def sum_even_numbers(n):
10     # Using the formula for the sum of the first k even numbers: k * (k + 1)
11     k = n // 2
12     return k * (k + 1)
13 # Test cases
14 print(sum_even_numbers(1000000)) # Should return 250000500000
15
16
17
```

The terminal at the bottom shows the output of the script:

```
400000.0
PS C:\Users\krish\OneDrive\Desktop\AIAC> & C:\Users\krish\AppData\Local\Programs\Python\Python313\p
ython.exe c:/Users/krish/OneDrive/Desktop/AIAC/week13.5/Task6.py
250000500000
PS C:\Users\krish\OneDrive\Desktop\AIAC>
```


EXPLANATION:

Loops over large ranges are slow; formulas like arithmetic progression or built-ins like `sum()` with comprehensions are faster. This reduces runtime complexity and improves efficiency significantly.

Task 7 (Removing Hidden Side Effects)

- **Task: Refactor code that modifies shared mutable state.**

- **Focus Areas:**

- o **Functional-style refactoring**

- o **Predictability**

Legacy Code:

```
data = []
```

```
def add_item(x):
```

```
    data.append(x)
```

```
add_item(10)
```

```
add_item(20)
```

```
print(data)
```

Expected Outcome:

- o **Refactored function returning values instead of mutating globals.**

PROMPT:

“Refactor functions that modify shared mutable state to return new values instead, following a functional style.”

CODE AND OUTPUT:

```
1  ...
2  data = []
3  def add_item(x):
4      data.append(x)
5      add_item(10)
6      add_item(20)
7      print(data)
8  ...
9  #Refactor functions that modify shared mutable state to return new values instead, following a
10 def add_item(data, x):
11     return data + [x]
12 # Test cases
13 data = []
14 data = add_item(data, 10)
15 data = add_item(data, 20)
16 print(data) # Should return [10, 20]
17
18
```

```
PS C:\Users\krish\OneDrive\Desktop\AIAC> & C:\Users\krish\AppData\Local\Programs\Python\Python313\python.exe c:/Users/krish/OneDrive/Desktop/AIAC/week13.5/Task7.py
[10, 20]
PS C:\Users\krish\OneDrive\Desktop\AIAC>
```

EXPLANATION:

This avoids hidden side effects by eliminating reliance on global or shared mutable data. Returning values makes behavior predictable and easier to test. It improves modularity and ensures functions remain pure, enhancing reliability.

Task 8 (Refactoring Complex Input Validation Logic)

- Task: Use AI to simplify and modularize complex validation rules.

• Focus Areas:

o Readability

o Testability

Legacy Code:

```
password = input("Enter password: ")
```

```
if len(password) >= 8:
```

```
if any(c.isdigit() for c in password):
```

```
if any(c.isupper() for c in password):
```

```
print("Valid Password")
```

else:

print("Must contain uppercase")

else:

print("Must contain digit")

else:

print("Password too short")

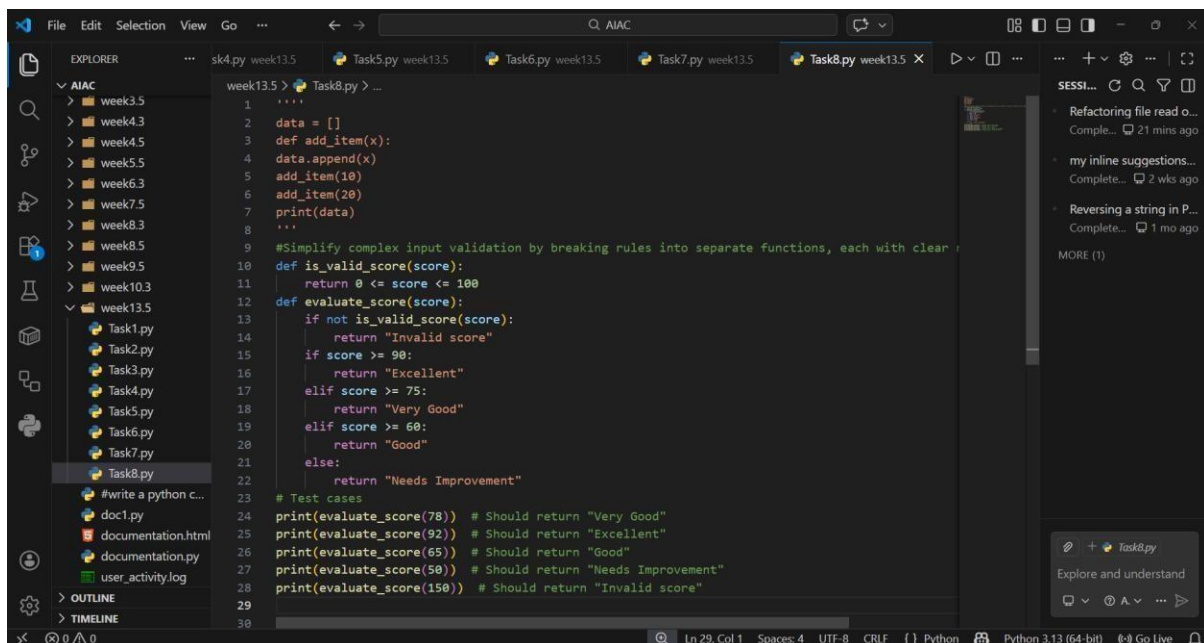
Expected Outcome:

o Separate validation functions with clear responsibility

PROMPT:

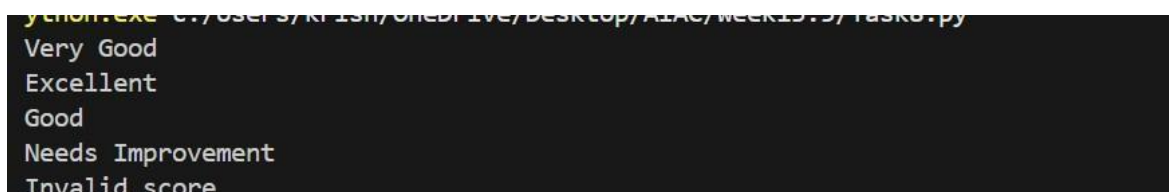
“Simplify complex input validation by breaking rules into separate functions, each with clear responsibility.”

CODE:



```
1  """
2  data = []
3  def add_item(x):
4      data.append(x)
5      add_item(10)
6      add_item(20)
7      print(data)
8  """
9  #Simplify complex input validation by breaking rules into separate functions, each with clear responsibility
10 def is_valid_score(score):
11     return 0 if 0 <= score <= 100 else 1
12 def evaluate_score(score):
13     if not is_valid_score(score):
14         return "Invalid score"
15     if score >= 90:
16         return "Excellent"
17     elif score >= 75:
18         return "Very Good"
19     elif score >= 60:
20         return "Good"
21     else:
22         return "Needs Improvement"
23 # Test cases
24 print(evaluate_score(78)) # Should return "Very Good"
25 print(evaluate_score(92)) # Should return "Excellent"
26 print(evaluate_score(65)) # Should return "Good"
27 print(evaluate_score(50)) # Should return "Needs Improvement"
28 print(evaluate_score(150)) # Should return "Invalid score"
29
30
```

OUTPUT:



```
python.exe C:/Users/RI/OneDrive/Desktop/AIAC/week13.5/Task8.py
Very Good
Excellent
Good
Needs Improvement
Invalid score
```

EXPLANATION:

Splitting validation into modular functions improves readability and testability. Each function handles one rule (length, digit, uppercase), making the logic easier to maintain. This approach supports unit testing and future extensions without nested conditionals.